

Customer Insights & Recommendation System

1. Project Overview

The Customer Insights & Recommendation Project aims to analyze customer purchasing behavior, identify key business patterns, segment customers, and generate product recommendations using data analytics techniques. An interactive Power BI dashboard was developed to provide business stakeholders with meaningful insights for data-driven decision-making.

2. Project Objectives

- Analyze customer purchasing behavior.
 - Identify top-performing products and regions.
 - Segment customers based on spending behavior.
 - Compare new and returning customers.
 - Generate product recommendations using association rule mining.
 - Build an interactive Power BI dashboard for business analysis.
-

3. Dataset Information

Dataset Used: Brazilian E-Commerce Public Dataset by Olist

The dataset contains information about customers, orders, products, payments, and reviews from a Brazilian e-commerce platform.

Files Used

- olist_customers_dataset.csv
- olist_orders_dataset.csv
- olist_order_items_dataset.csv
- olist_products_dataset.csv
- olist_order_payments_dataset.csv
- olist_order_reviews_dataset.csv
- product_category_name_translation.csv

The dataset contains approximately 100,000+ records.

4. Data Preprocessing

Several preprocessing steps were performed before analysis:

Data Cleaning

- Removed duplicate records.
- Filtered only delivered orders.
- Handled missing values using median and mode techniques.
- Converted date columns into datetime format.

Data Merging

Multiple datasets were merged using common keys such as:

- customer_id
- order_id
- product_id

Feature Engineering

New features were created:

- Year
- Month
- Day
- Order_Value
- Delivery_Days

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify trends and patterns.

Analyses Performed

- Monthly Sales Trend Analysis
- Top Product Category Analysis
- State-wise Sales Analysis
- Payment Method Distribution
- Review Score Analysis
- Customer Segmentation
- Top Customer Identification
- Correlation Analysis

Several visualizations were generated using Matplotlib and Seaborn.

6. Customer Segmentation

Customers were segmented based on their total spending.

Three customer groups were identified:

- High Value Customers
- Medium Value Customers
- Low Value Customers

Segmentation helps businesses target customers more effectively and improve marketing strategies.

7. Recommendation System

A product recommendation system was developed using:

Technique Used

Association Rule Mining (Apriori Algorithm)

The recommendation system identifies products that are frequently purchased together.

Important metrics used:

- Support
- Confidence
- Lift

These metrics help in understanding customer buying patterns and cross-selling opportunities.

8. Power BI Dashboard

An interactive dashboard was developed in Power BI consisting of three pages.

Page 1: Executive Overview

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value
- Average Delivery Time
- Monthly Sales Trend
- Product Performance
- Payment Method Distribution

- Sales by State Map

Page 2: Customer Insights

- Customer Segmentation
- Top Customers
- New vs Returning Customers
- Business Recommendations
- Product Recommendation Table

Page 3: Advanced Analytics

- Review Score Analysis
- Delivery Performance Analysis
- State-wise Analysis
- Monthly Order Trend
- Interactive Filters and Slicers

9. Streamlit Web Application

In addition to the Power BI dashboard, an interactive web application was developed using Streamlit.

Features

- KPI Cards
- Interactive Charts
- Customer Analytics
- Product Recommendations
- Sidebar Filters
- Dataset Preview
- Downloadable Data

The Streamlit application enables users to interact with the data dynamically and explore business insights in real time.

10. Key Insights

- Credit Card is the most preferred payment method.
- Bed Bath Table category generated the highest revenue.
- A small number of customers contribute significantly to overall revenue.

- High-value customers represent the most important customer segment.
 - Certain states contribute more revenue compared to others.
 - Average customer review score is approximately 4 out of 5.
 - Average delivery time is approximately 12 days.
-

11. Business Recommendations

1. Focus marketing efforts on high-value customers.
 2. Improve customer retention strategies for repeat purchases.
 3. Promote top-selling product categories.
 4. Optimize delivery performance to improve customer satisfaction.
 5. Increase marketing campaigns in top-performing states.
 6. Use recommendation systems for cross-selling opportunities.
-

12. Conclusion

The project successfully transformed raw e-commerce data into actionable business insights through data preprocessing, exploratory analysis, customer segmentation, and recommendation systems.

Interactive dashboards developed using Power BI and Streamlit enable stakeholders to monitor business performance, analyze customer behavior, and support data-driven decision-making.

This project demonstrates practical applications of Data Analytics, Business Intelligence, Customer Segmentation, Recommendation Systems, and Interactive Data Visualization techniques in solving real-world business problems.